BBO capstone project presentation template

Develop a concise presentation that captures:

1. **An overview of your BBO approach**
   - Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
   - Briefly describe your overall strategy and how your process fits together.

OBJECTIVE

The project aims to find strong input settings for eight unknown functions while using as few expensive evaluations as possible. Each function accepts between two and eight continuous inputs and returns one score. All eight tasks are maximisation problems. The equations, gradients and global optima are hidden, so every query must either improve the best observed score or reduce uncertainty about the next decision.

OVERALL APPROACH

1. Record the raw query and returned portal value for each function.

2. Fit a separate surrogate and review local response patterns. Early rounds used Gaussian Processes with Upper Confidence Bound and Expected Improvement. Later checks included regression, SVMs and small neural-network ensembles.

3. Generate candidate points through global sampling or small local movements around strong observations.

4. Compare predicted performance, uncertainty, distance from previous points and recent empirical trends.

5. Validate the chosen vector for bounds, dimensionality, six-decimal formatting and duplication before submission.

The process is iterative. Each returned result is appended to the history, the model is refitted and the exploration-exploitation balance is reconsidered separately for each function. Through Week 11, the evidence base contains 263 completed evaluations, including the supplied initial data.

2. How your strategy has evolved

- Describe the key changes in your approach since your early rounds.
- What influenced these changes – data trends, feedback, model performance or intuition?
- Identify which principles or heuristics now guide your query decisions.

EARLY ROUNDS

I began with one broad Bayesian optimisation policy. Separate Gaussian Processes estimated the expected output and uncertainty for each function. Upper Confidence Bound favoured candidates with a strong predicted value, high uncertainty or both. Expected Improvement and Sobol candidate generation provided sensitivity checks.

MIDDLE ROUNDS

The returned outputs showed that one acquisition setting did not suit all eight functions. I introduced function-specific exploration levels and compared interpretable regression, nonlinear SVM and neural-surrogate perspectives. These models helped identify local feature sensitivity, but sparse data made complex predictions unstable.

LATER ROUNDS

Repeated observations shifted the process toward transparent local rules. I used bracketing, interpolation, isolated coordinate changes and controlled continuation along directions that improved over several rounds. Cluster medoids helped prevent isolated poor points from moving the search centre. PCA-style reasoning highlighted repeated displacement vectors and redundant coordinates.

CURRENT PRINCIPLES

- Preserve raw observations separately from predictions.
- Continue a direction only after repeated improvement.
- Shrink, reverse or bisect a move after deterioration.
- Change one coordinate at a time when attribution is unclear.
- Use uncertainty for diagnostic exploration, not as automatic evidence of quality.
- Prefer reproducible rules and small reversible steps near strong incumbents.

The changes were driven mainly by observed trends and model errors. Intuition helped propose alternatives, but portal results determined whether a strategy was retained.

3. Patterns, data and insights

- Highlight the most meaningful trends you've observed across your data points or model evaluations.
- Identify which variables or behaviours seem to influence results the most.
- Explain how these observations have shaped your understanding of the search process.

MEANINGFUL PATTERNS

All eight functions improved on their best initial observation, but their local structures differ.

- Function 1 has a compact peak near both coordinates equal to 0.58. Small distance from this region matters more than large independent movements.
- Function 2 performs best near x2 = 1 and x1 around 0.681, but very close points can return sharply different values. Noise or a narrow unresolved feature appears important.
- Function 3 has a narrow strong region near [0.518, 0.605, 0.424]. Mirrored and midpoint probes have not improved the Week 6 incumbent.
- Function 4 follows a stable coordinated direction near [0.414, 0.416, 0.353, 0.426]. Successive short movements produced new incumbents.
- Function 5 shows the clearest feature effect. With x1, x3 and x4 fixed on a boundary face, increasing x2 raised the best output from 1088.86 initially to 1623.35 by Week 11.
- Function 6 is most sensitive locally to x4, although nearby results remain inconsistent.
- Function 7 follows a repeatable six-variable direction and reached 1.81327 in Week 11.
- Function 8 follows a boundary ridge, but recent gains flattened and then reversed.

INTERPRETATION

The most useful signals are local directions, boundary behaviour and repeated improvement rather than universal feature importance. Input distance alone is insufficient because Functions 2 and 6 show high local output variation. Later queries therefore combine proximity with output rank, stability and direction of travel. Many nearby observations are also redundant, so the remaining budget should focus on movements that add distinct evidence.

4. Decision-making and iteration

- Explain how you balance exploration and exploitation.
- Describe one or two examples of strategic decisions. What worked, what didn't, and why?
- Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

BALANCING EXPLORATION AND EXPLOITATION

Early rounds gave more weight to exploration because the available observations covered little of each search space. As credible high-performing regions emerged, exploitation became more important. I still retain small exploratory probes when they distinguish between explanations, such as whether a poor result reflects noise, a narrow peak or an incorrect direction.

WHAT WORKED: FUNCTION 5

A controlled sequence varied only x2 while keeping x1, x3 and x4 fixed at [0, 1, 1]. Increasing x2 in regular 0.025 steps repeatedly improved the output. This reduced a four-dimensional local problem to one active direction and made each result easy to interpret. The best observed output increased from 1088.86 in the initial data to 1623.35 in Week 11.

WHAT REQUIRED ADAPTATION: FUNCTION 2

Function 2 returned very different outputs for extremely similar inputs near x2 = 1. A simple smooth curve and fixed continuation steps therefore became unreliable. I responded by keeping the strong boundary coordinate fixed, reducing movements in x1 and treating the Week 9 incumbent as a medoid rather than allowing neighbouring poor results to shift the centre.

HANDLING UNEXPECTED RESULTS

When a query underperforms, I do not treat it only as failure. I ask whether it rejects a direction, reveals noise or provides a useful boundary. I then refit the surrogate, compare simpler local trends, reduce the step size and separate predictions from observations. No single model decides the query when models disagree. This approach limits overreaction to one value while still allowing the strategy to change when repeated evidence contradicts an assumption.

5. Next steps and reflection

- Outline your planned actions or experiments to improve the model's performance.
- Include two or three sentences connecting your work to the broader ML landscape.
- Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

PLANNED ACTIONS

The remaining rounds will test whether the strongest empirical directions are durable. Functions 4, 5 and 7 will receive another measured continuation only if Week 12 confirms improvement. Functions 1, 3 and 6 will stay close to established incumbents with smaller neighbourhood probes. Function 2 will use conservative one-coordinate changes because its local response remains noisy. Function 8 will test a reduced or reversed move after its latest continuation regressed.

For the final round, I will compare each new result with the incumbent and choose between one last continuation and a protective local refinement. I will also update the query log, README, datasheet and model card so the final evidence and assumptions remain reproducible.

CONNECTION TO MACHINE LEARNING

This project resembles active learning and experimental design because the optimiser chooses which expensive observation to purchase next. It also illustrates a wider ML lesson: model flexibility helps only when the data support it. With sparse adaptive samples, uncertainty estimates, validation and transparent documentation matter as much as predictive fit.

The workflow also reflects responsible model development. Raw observations remain separate from predictions, unsuccessful experiments stay in the record and claims are limited to observed improvement. The unknown global optima prevent claims that the best possible settings have been found.

STAKEHOLDER MESSAGE

I would explain the result in plain terms: I tested a limited number of carefully selected settings, learned which regions performed consistently better and narrowed the search around them. Every function improved on its starting benchmark, although some results remain noisy and the project cannot prove that it found the global best setting.